MATH 10A — STRUCTURE AND CONCEPTS IN MATH I

§2.2

Models and Strategies for Addition and Subtraction

Many ways to add and subtract, and how to choose one

Today's Goal

After completing this section, you should be able to:

- Use the set, number line, and missing-addend models to represent addition and subtraction of whole numbers
- Apply multiple strategies, including the standard algorithms, to add and subtract whole numbers using appropriate representations

LT.4 Addition and Subtraction Strategies for Whole Numbers

I can apply multiple strategies (including the standard algorithms) for addition and subtraction of whole numbers using appropriate representations.

Two Children, One Sum

Two second graders were asked to find $58 + 27$. Here is what each one wrote.

Child A

$60 + 25 = 85$

Child B

$50 + 20 = 70$

$8 + 7 = 15$

$70 + 15 = 85$

Is each child right? What did each child do? Talk with your shoulder partner.

Both are right. Child A moved 2 from 27 to 58 to make a friendly number: that is **opposite-change**. Child B added the tens and the ones separately: that is **partial sums**.

Three Models Behind Every Strategy

- **Set model.** A set of objects stands for each quantity. Addition combines two sets. Subtraction removes a subset. This is the model behind direct modeling with cubes.
- **Number line model.** A directed arrow stands for each quantity. The length of the arrow is the number of unit segments it covers, not the number of tick marks it touches.
- **Missing addend model.** Subtraction is reframed as addition: $a - b$ asks what must be added to b to reach a.

Subtraction Is a Missing Addend

The missing addend model matters because it ties subtraction to addition, and it is often used to define subtraction.

Definition 2.2.12

For any two whole numbers a and b, where $a \geq b$, $a - b$ is defined to be the unique whole number c such that $b + c = a$.

Keep this in mind. One of today's subtraction strategies is this definition, used directly.

Twelve Strategies, Five Today

Addition

- Direct modeling
- Counting on
- Doubles
- Making 10 and friendly numbers TODAY
- Standard algorithm
- Partial sums
- Opposite-change TODAY

Subtraction

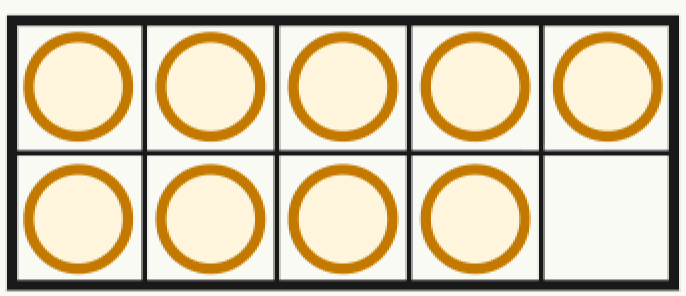
- Counting up TODAY
- Standard algorithm TODAY
- Trades-first
- Left-to-right
- Same-change TODAY

The other seven are worked out in the notes. The Prep and Practice cover them.

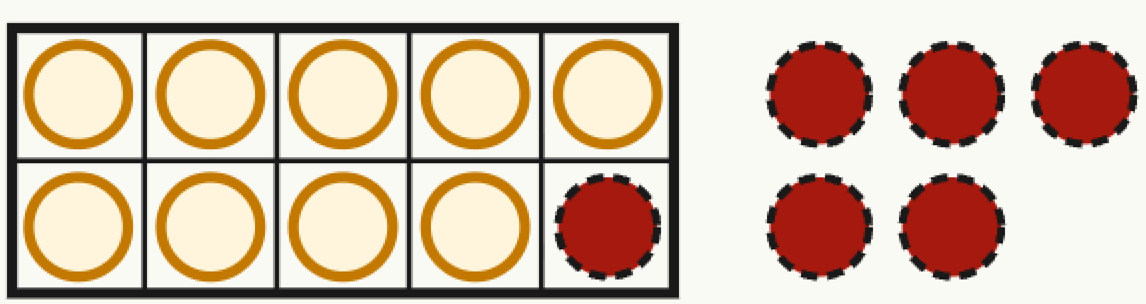
Making 10 With a Ten Frame

Use the making 10 strategy to evaluate $9 + 6$.

Start with 9 light counters.



Add 6 dark: 1 fills the frame, 5 stay outside.



The same move as a diagram: split 6 into $1 + 5$.

$$9 + 6$$

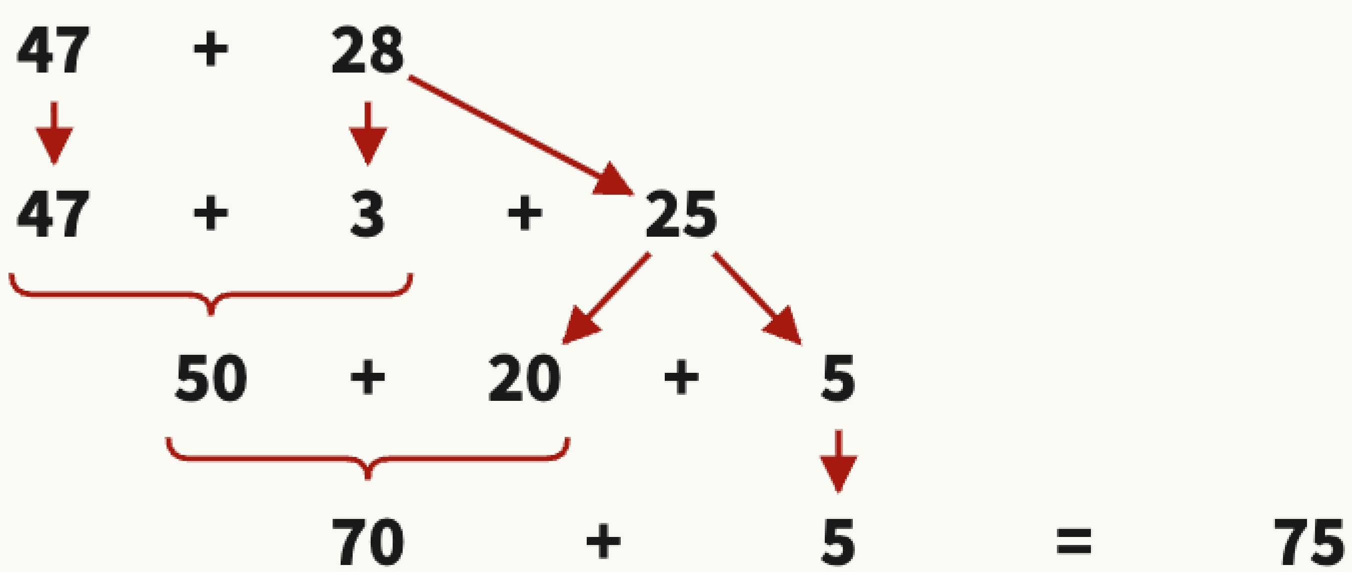
$$\underbrace{9 + 1}_{10} + 5$$

$$10 + 5 = 15$$

The full frame counts 10. With the 5 outside, $9 + 6 = 15$.

Making a Friendly Number

Use a friendly number to evaluate $47 + 28$.



47 needs 3 to make 50. Split 28 into $3 + 25$.

$47 + 3 = 50$. Split the leftover 25 into $20 + 5$ and add the tens first.

$50 + 20 = 70$, then $70 + 5 = 75$.

Either addend can be the one you complete: 28 needs 2 to make 30. Go to the **nearest** ten. A small change is easier to compensate.

Two Changes, Opposite Directions

Use the opposite-change algorithm to evaluate $364 + 297$.

$$
\begin{array}{rcr}
364 & \xrightarrow{-3} & 361 \\
+\quad 297 & \xrightarrow[+3]{} & +\quad 300 \\
\hline
 & & 661
\end{array}
$$

Which addend is close to a friendly number? 297 is 3 from 300. Add 3.

To compensate, subtract 3 from 364: 361.

Now add: $361 + 300 = 661$.

The total does not change: what one addend gains, the other loses. This is also called **compensation**.

Two Ways for Every Sum

Find your group's sum. Show it **two ways**: the standard algorithm, and one mental strategy from the map.

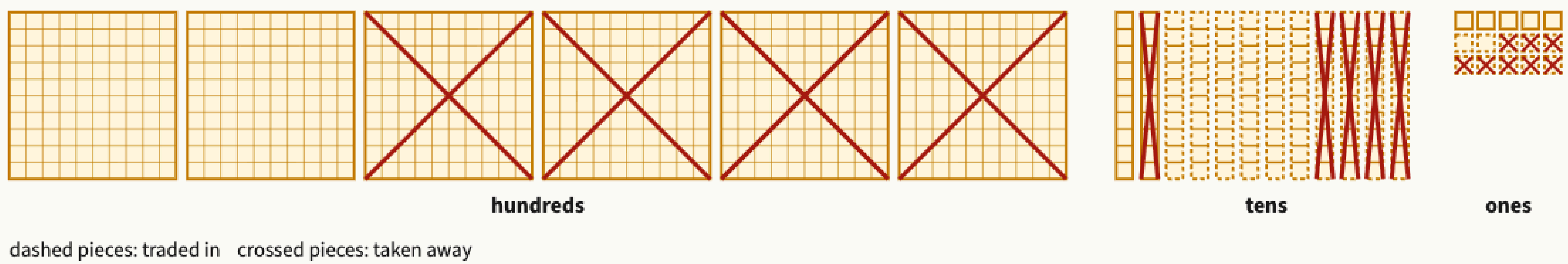
Groups	Sum
$1, 5, 9$	(a) $199 + 456$
$2, 6$	(b) $3487 + 2615$
$3, 7$	(c) $68 + 57$
$4, 8$	(d) $745 + 138$

Then decide: which way was faster, and would that change for a different sum?

Answers: (a) 655 (b) 6102 (c) 125 (d) 883

Regrouping, One Place at a Time

Use the standard algorithm to evaluate $625 - 348$.



$$\begin{array}{cccc}
 & & 11 & \\
 & 5 & \cancel{1} & 15 \\
 & \cancel{6} & \cancel{2} & \cancel{5} \\
- & 3 & 4 & 8 \\
\hline
 & 2 & 7 & 7
\end{array}$$

Ones: trade 1 ten for 10 ones.

$15 - 8 = 7$ ones.

Tens: trade 1 hundred for 10 tens.

$11 - 4 = 7$ tens.

Hundreds: $5 - 3 = 2$. So $625 - 348 = 277$.

Shift Both, Keep the Difference

Use the same-change algorithm to evaluate $734 - 486$.

$$\begin{array}{rcrcr} 734 & \xrightarrow{+4} & 738 & \xrightarrow{+10} & 748 \\ -\ 486 & \xrightarrow[+4]{} & -\ 490 & \xrightarrow[+10]{} & -\ 500 \\ \hline & & & & 248 \end{array}$$

Make the subtrahend end in zero. 486 needs 4 to reach 490. Add 4 to both.

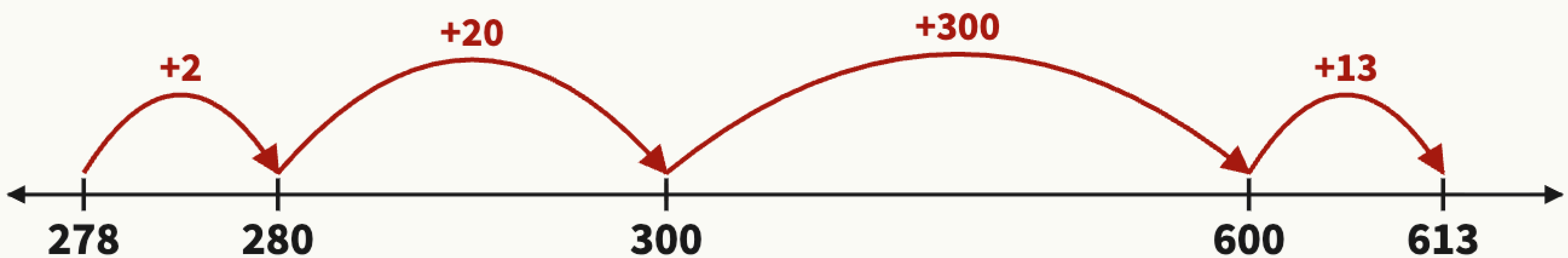
Make it end in two zeros. 490 needs 10 to reach 500. Add 10 to both.

Now subtract: $748 - 500 = 248$.

Both numbers moved the same way, so the gap between them did not change. In one move: add 14 to both.

Count Up From the Subtrahend

Use the counting up algorithm to evaluate $613 - 278$.



$+2$ to the next ten: 280.

$+20$ to the next hundred: 300.

$+300$ to the last hundred before 613: 600.

$+13$ for the rest: 613.

Add the jumps:
$2 + 20 + 300 + 13 = 335$. So $613 - 278 = 335$.

This is the missing addend model at work: $278 + \square = 613$.

Which Strategy Fits These Numbers?

Group member	Difference
1	(a) $7000 - 3568$
2	(b) $802 - 795$
3	(c) $968 - 423$
4	(d) $641 - 279$

a. Work your own difference. Choose the strategy you think is most efficient and show how it would be used.

b. Take turns. Show your group what you did. Did someone have a more efficient strategy?

Answers: (a) 3432 (b) 7 (c) 545 (d) 362

One Question to Leave With

A second grader must find $62 - 38$. Which strategy would you teach first?

What must the child already know for it to work?

More than one answer is good. A strong answer names what the child must know: friendly tens and the missing addend for counting up, regrouping for the standard algorithm, the fixed gap for same-change.